AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 33 · WORKBOOK

Model Cards and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable model-card and review record.

Audit arithmetic,correct unsupported claims and request missing evidence.

Keep one habit: W1–W3/W5 → selected newV2claim → actual project card → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

Proxy Features and Fairness

> **My remembered idea, example and correction**

Improve Generalization

> **My remembered idea, example and correction**

Overfitting

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main paperV1 card:fictional parcel-lane clear/blocked exercise, training 40/validation 20/test 12 separate source-group premise. TestTN7/FP 1/FN 2/TP 2 givesaccuracy9/12, precision 2/3, recall 2/4; fixed-clearbaseline 8/12. CompleteR01–R12 saved pairs audit arithmetic, not missing training/code artifacts. PictureV3 instead16/20 photosfrom5 unseenplants×4 views, indoor tablet, baseline 14/20.

  No class matrix or recall supplied forV3. A card preserves unknowns, excluded uses and pending review; it is not certification or a completed sign-off.

- Reference for this exercise: open this week's student lesson, "Model Cards and Human Review", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Write an intended-use and non-use statement for the worked card.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main paperV1 card:fictional parcel-lane clear/blocked exercise, training 40/validation 20/test 12 separate source-group premise. TestTN7/FP 1/FN 2/TP 2 givesaccuracy9/12, precision 2/3, recall 2/4; fixed-clearbaseline 8/12. CompleteR01–R12 saved pairs audit arithmetic, not missing training/code artifacts. PictureV3 instead16/20 photosfrom5 unseenplants×4 views, indoor tablet, baseline 14/20.

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- Reference for this exercise: open this week's student lesson, "Model Cards and Human Review", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Verify the worked accuracy, precision and recall.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main paperV1 card:fictional parcel-lane clear/blocked exercise, training 40/validation 20/test 12 separate source-group premise. TestTN7/FP 1/FN 2/TP 2 givesaccuracy9/12, precision 2/3, recall 2/4; fixed-clearbaseline 8/12. CompleteR01–R12 saved pairs audit arithmetic, not missing training/code artifacts. PictureV3 instead16/20 photosfrom5 unseenplants×4 views, indoor tablet, baseline 14/20.

  No class matrix or recall supplied forV3. A card preserves unknowns, excluded uses and pending review; it is not certification or a completed sign-off.

- A card says fair and reliable but provides only accuracy. Request three additions.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main paperV1 card:fictional parcel-lane clear/blocked exercise, training 40/validation 20/test 12 separate source-group premise. TestTN7/FP 1/FN 2/TP 2 givesaccuracy9/12, precision 2/3, recall 2/4; fixed-clearbaseline 8/12. CompleteR01–R12 saved pairs audit arithmetic, not missing training/code artifacts. PictureV3 instead16/20 photosfrom5 unseenplants×4 views, indoor tablet, baseline 14/20.

  No class matrix or recall supplied forV3. A card preserves unknowns, excluded uses and pending review; it is not certification or a completed sign-off.

- What should change in a card after a threshold revision?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- The selectedV2 card supplies only "90%accurate"and "works for every school." It gives no numerator, denominator, source context, split history, baseline or predictions. You cannot audit the 90%or replace it with a new invented score. Request the missing evidence and write a bounded unknown-scope statement while preserving the unverified reported claim as unverified.

- Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Complete the Plant Classifier **model card** and require teacher sign-off before any classroom demonstration.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Check arithmetic for **Model card**s and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Model card**s and review. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A **model card** says healthy/spotted,**80%** accuracy only.
- Add responsible-use **evidence**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A **model card** says 90% accuracy but gives no **intended use**, data conditions or human-review rule. What should be added before classroom use?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 33 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Write a **model card** whose performance statement can be traced to a concrete evaluation.
- Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored14/20.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/33

Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name evidence

Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.

2. Compare baseline

v = 3 accuracy 80%; baseline70% on the same 20 cases.

3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Write a **model card** whose performance statement can be traced to a concrete evaluation.
- Which labelled step matches this detail? Do not claim all cameras, outdoor conditions or disease diagnosis were validated.
- 1. Name **evidence**
- 2. State limits
- 3. Compare baseline
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State limits Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Read the picture in order.

Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The publicity sentence becomes "**80%** accurate for every plant". Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Reject that unbounded statement.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Repair and keep the record

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

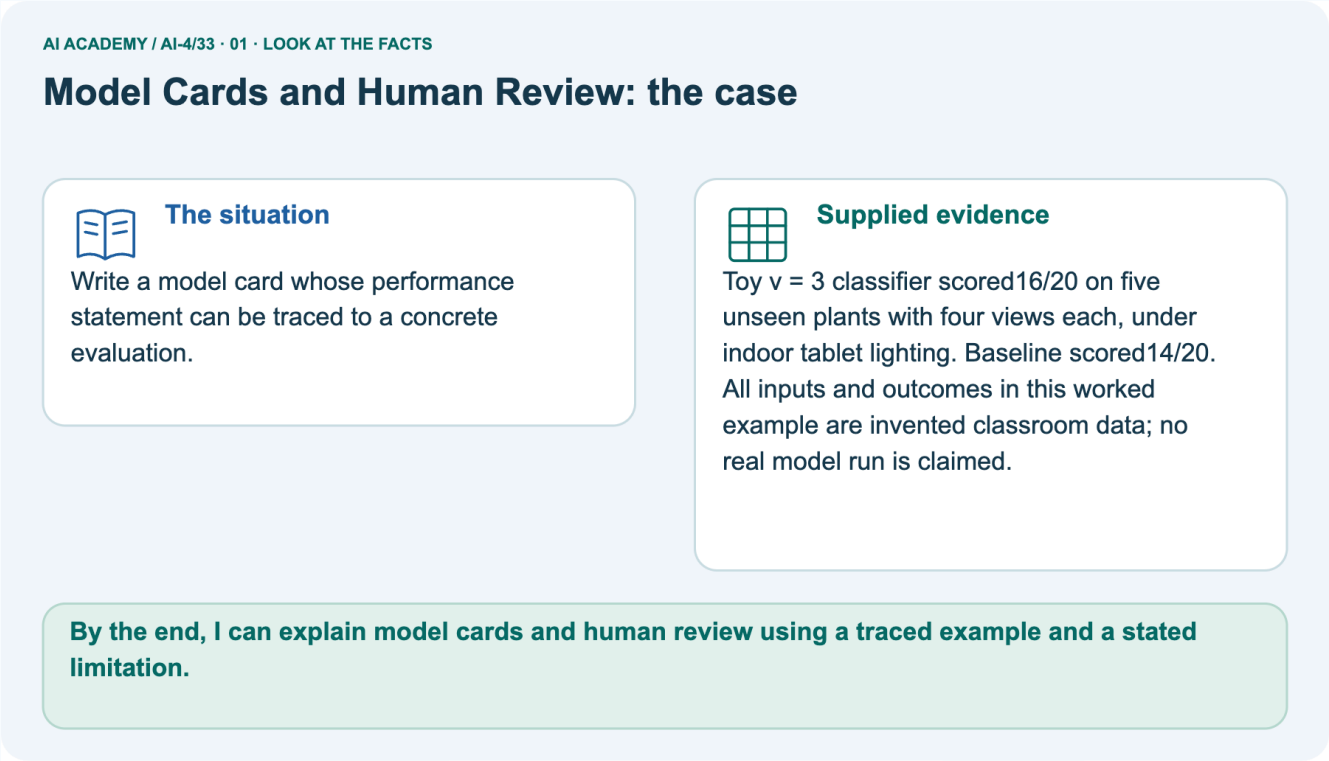
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Model Cards and Human Review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

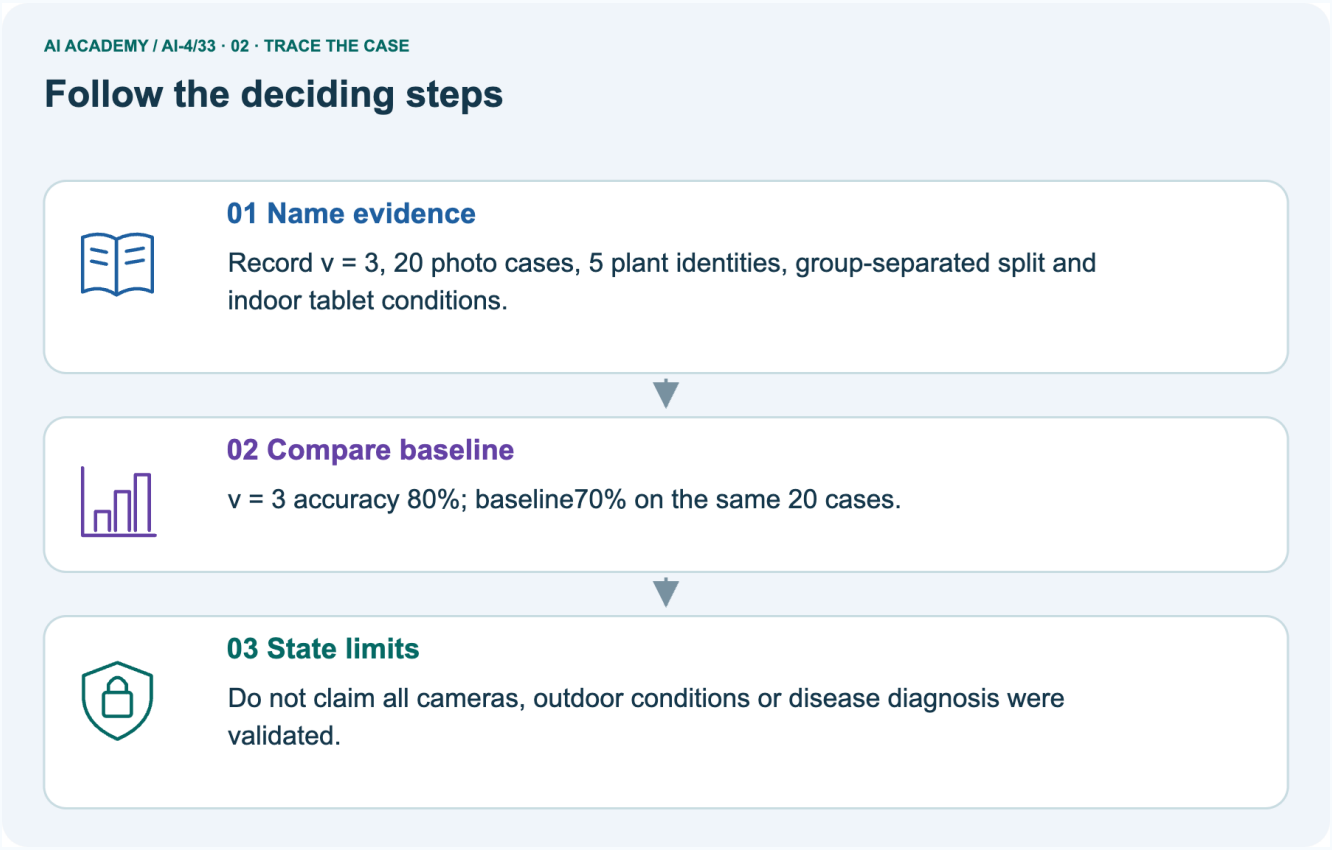
Write a **model card** whose performance statement can be traced to a concrete evaluation. Supplied facts: Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting.

Baseline scored14/20. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Name **evidence**: Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions. → Compare baseline: v = 3 accuracy **80%**; baseline**70%** on the same 20 cases. → State limits: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Model cards and review at a glance

AI ACADEMY / AI-4/33 · 03 · INSPECT THE LESSON MODEL

Model cards and review at a glance

Part	Evidence or action	Interpretation
Purpose and version	Toy classifier v 1	Fictional data only
Evidence	12 held-out cases	Full matrix and baseline
Limits and review	No real decisions	List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

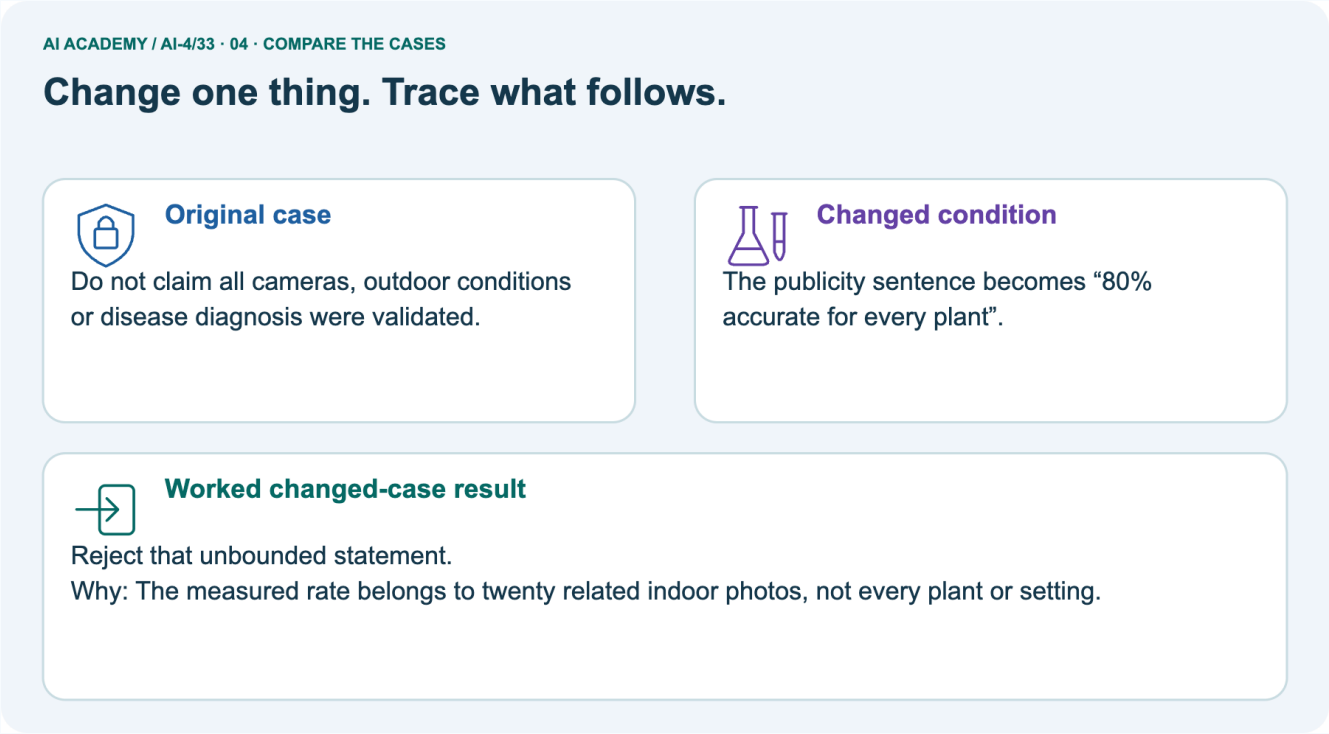
Follow the labelled picture with the matching explanation. Open the vector diagram

Model cards and review at a glance. Purpose and version | Toy classifier v 1 | Fictional data only. **Evidence** | 12 held-out cases | Full matrix and baseline. Limits and review | No real decisions | List unresolved coverage

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

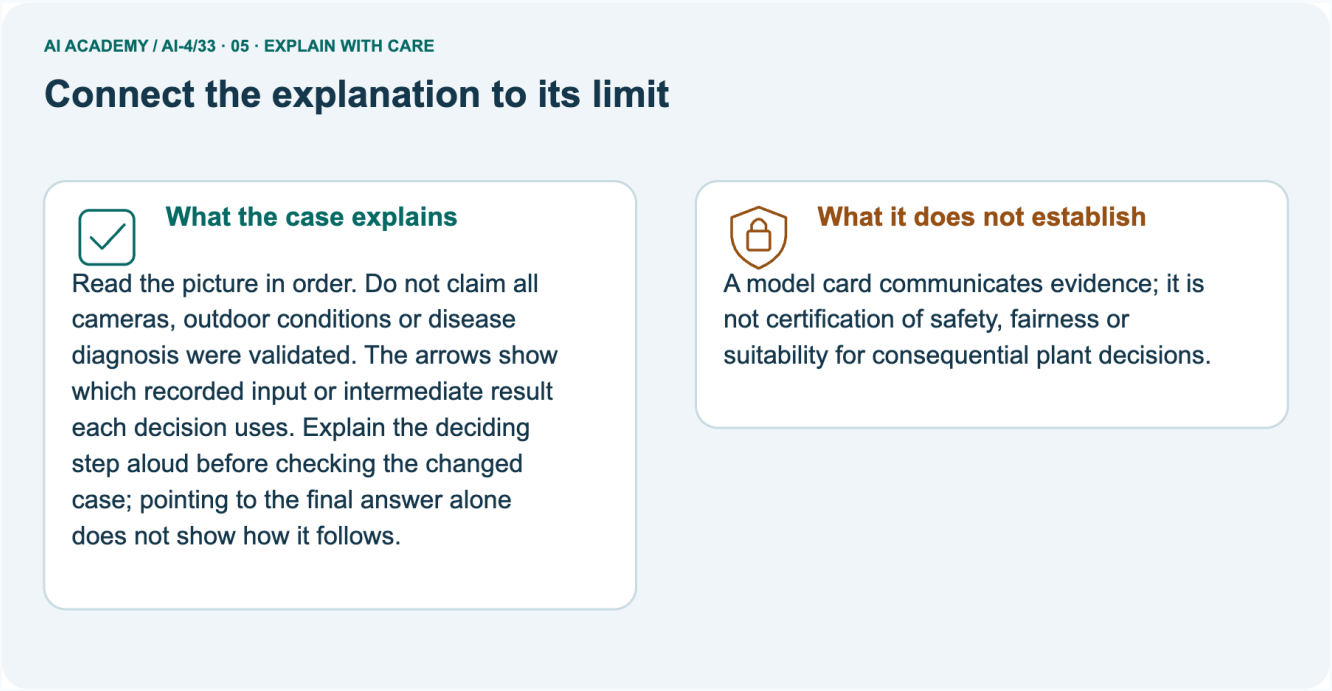
Original case: Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Changed condition: The publicity sentence becomes “**80%** accurate for every plant”.

Expected result: Reject that unbounded statement. Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

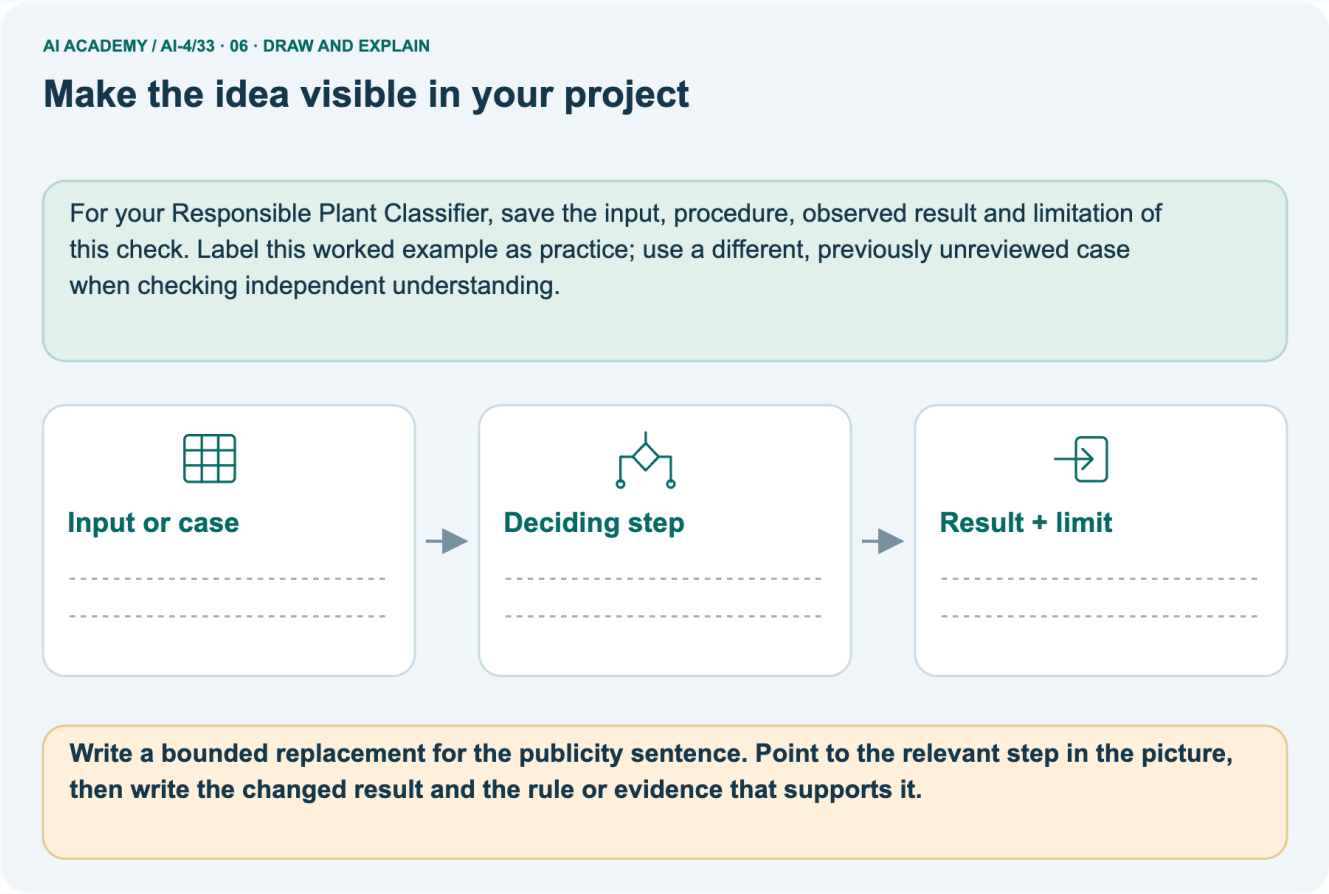
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Model Cards and Human Review: the case

02. Follow the deciding steps

03. Model cards and review at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.